

EXAM 7 – SPRING 2011

6. (2.75 points)

Given the following information for accident year 2009 workers compensation business with a \$250,000 deductible as of December 31, 2010:

	<u>Ground Up Reported Losses</u>	<u>Ground Up Reported Severity</u>	<u>24 Months to Ultimate LDF</u>
Limited to \$250,000	\$4,000,000	\$125,000	1.25
Unlimited	4,500,000	140,000	1.50

- Full coverage premium for all policies: \$12,500,000
- Full coverage expected loss ratio: 0.700
- Excess ratio (per-occurrence charge): 0.150
- Excess ratio (aggregate charge): 0.020

a. (1.25 points)

Estimate the IBNR for losses above the deductible using the direct development method.

b. (1.25 points)

Estimate the IBNR for losses above the deductible using the Bornhuetter-Ferguson method.

c. (0.25 point)

Briefly describe an advantage of using the Bornhuetter-Ferguson method to estimate IBNR for workers compensation high deductible business.

CONTINUED ON NEXT PAGE

Sample Solutions
Exam 07, 2011, Question 6

Solution 1

a)

Excess losses = Unlimited losses – Limited losses = \$4.5 M - \$4.0 M = \$0.5 M

Find Excess LDF : Unlimited loss x Unlimited LDF = Limited loss x Limited LDF +

Excess loss × Excess LDF

\$4.5 M x 1.5 = \$4 M × 1.25 + \$0.5 M × Excess LDF

Excess LDF = 3.5

Excess IBNR = (Reported Excess losses) × (Excess LDF -1) = \$0.5 M × (3.5 – 1) = \$1.25 M

b)

$IBNR_{BF} = \text{Expected loss} \times \text{unreported \%} ; \text{unreported \%} \text{ xs loss}$

$= 1 - 1/3.583 = 72.093\%$

Expected xs loss = $12.5 \text{ M} \times .7 \times .15 + 12.5 \times .7 \times (1-.15) \times .02 = 1.46125 \text{ M}$

$IBNR_{BF} = \text{Expected xs loss} \times (1 - 1/3.583) = 1.46125 \text{ M} \times .72093 = 1,053,459$

c) Advantage of B/F method -> is more stable overall and can be tied to pricing estimates